

Khaja Mujahiddin

New Haven, CT (Open to Relocate) | 347-736-5812 | khajamujahiddin@gmail.com | [Portfolio](#) | [Github](#)

Data Engineer

SUMMARY

Senior Data Engineer with 5+ years of experience architecting scalable ELT pipelines, Medallion Lakehouse infrastructures, and real-time streaming systems. Proven expertise within the Databricks, Snowflake, and Azure ecosystems, driving high-impact engineering initiatives in healthcare and retail domains. Adept at transitioning legacy batch processes into highly available, event-driven architecture and building secure, PHI-compliant data supply chains to support Agentic AI integrations, predictive modeling, and enterprise analytics.

TECHNICAL SKILLS

- **Cloud & Infrastructure:** Azure (Data Factory, ADLS Gen2, Event Hub, Functions), AWS (S3, Lambda, Redshift).
- **Data Engineering & Processing:** Databricks (Delta Lake, Unity Catalog), PySpark, Apache Spark, Snowflake, dbt (Data Build Tool), ETL/ELT pipelines.
- **Real-Time & Streaming:** Apache Kafka, Azure Event Hub, Event-Driven Architectures.
- **Databases:** SQL (Advanced), PostgreSQL, MySQL, NoSQL.
- **Orchestration & DevOps:** Apache Airflow, Databricks Workflows, Docker, Kubernetes, Git, GitHub Actions, CI/CD.
- **Languages & Interoperability:** Python, SQL, REST APIs, FHIR, HL7.

PROFESSIONAL EXPERIENCE

Optum | Hartford, CT

Jul 2025 – Present

Data Engineer

- Architected secure, PHI-compliant data pipelines utilizing Azure Databricks and PySpark to support the launch of Digital Auth Complete, directly contributing to a 45% reduction in manual administrative touches and a projected 96% first-pass prior authorization rate.
- Engineered high-performance Python APIs and deployed serverless Azure Functions to extract structured and unstructured clinical documentation from over 250 disparate payer systems and Electronic Health Records (EHR).
- Ensured all external health data ingestion adhered strictly to FHIR and HL7 interoperability standards to seamlessly feed automated clinical AI review models.
- Designed and implemented a governed Medallion Lakehouse architecture (Bronze, Silver, Gold layers) using Delta Live Tables and Spark SQL.
- Partnered with Data Science squads to build secure Retrieval-Augmented Generation (RAG) data supply chains integrating curated clinical datasets with enterprise LLMs.
- Enforced HIPAA-compliant governance and fine-grained access controls using Databricks Unity Catalog.
- Maintained robust CI/CD pipelines using GitHub Actions for automated testing and reliable deployments.

Staples | Chennai, IN

Oct 2021 – Jul 2023

Jr. Data Engineer

- Modernized Enterprise Merchandising & Pricing Technology backend by migrating legacy batch workflows to scalable ELT architectures utilizing Snowflake and Databricks, reducing data latency by 40%.
- Developed modular SQL transformations utilizing dbt to model complex discount lifecycles and pricing agreements.
- Architected event-driven pipelines utilizing Apache Kafka and Azure Event Hub for real-time synchronization of pricing calculations.
- Optimized PostgreSQL schemas and advanced SQL queries to reduce latency and accelerate demand-forecasting algorithms.
- Automated multi-stage workflows utilizing Apache Airflow and Databricks Workflows.
- Implemented CI/CD best practices via Azure DevOps to maintain high code quality and operational continuity.

PowerBI Developer

- Developed interactive data visualization dashboards using Power BI and Tableau for enterprise consulting clients, converting complex network telemetry and operational metrics into actionable business intelligence.
- Authored and optimized advanced SQL queries for data extraction and transformation of historical service logs, implementing strict data quality checks to ensure 100% data integrity for downstream reporting.
- Engineered text-mining and data preprocessing pipelines in Python to ingest, clean, and structure over 100,000 monthly inbound customer IT support tickets for NLP tokenization and intent classifiers.
- Collaborated on predictive modeling initiatives with senior data scientists, leveraging statistical analysis on network data to identify client infrastructure bottlenecks.
- Executed cross-functional requirement gathering by working directly with business stakeholders to translate ambiguous client problems into technical data models and reporting specifications.

ACADEMIC & APPLIED RESEARCH PROJECTS

Agentic AI Automated Clinical Workflow

- Built a multi-step Agentic AI data pipeline using Python, LangChain, and OpenAI APIs to autonomously ingest and summarize simulated clinical notes.
- Implemented a Retrieval-Augmented Generation (RAG) backend utilizing Pinecone for semantic search and reduced hallucinations.

Multimodal Fake News Detection System

- Engineered a distributed data ingestion pipeline to process unstructured text and image datasets from social media feeds.
- Developed NLP and Computer Vision feature extraction workflows achieving 92% classification accuracy.

EDUCATION

University of New Haven | West Haven, CT

Aug 2023 – May 2025

Master of Science, Data Science